

HANEDA AP, Japan

WMO: 476710

Lat: 35.553N

Lon: 139.781E

Elev: 31

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.7	35.5	8.6	8.6	39.6	12.0	10.2	40.3	29.6	47.0	27.4	45.9	11.9	330	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	91.3	78.9	89.5	78.0	87.6	77.7	80.3	86.9	79.5	85.9	78.7	84.8	15.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	149.6	84.8	77.3	142.2	83.7	76.8	139.8	83.4	44.1	88.2	43.1	86.0	42.2	85.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	25.0	22.5	30.1	95.3	2.0	2.5	28.6	97.1	27.5	98.5	26.4	99.9	25.0	101.7
			24.3	81.9	1.8	1.7	23.0	83.1	21.9	84.1	20.9	85.0	19.6	86.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	43.3	44.2	49.7	58.4	66.6	72.0	79.3	82.0	75.6	66.4	56.6	47.9
	DBStd	13.99	3.68	4.49	5.59	5.61	4.63	4.31	4.83	4.28	5.40	4.77	4.77	4.41
	HDD50	560	210	172	75	6	0	0	0	0	0	3	94	
	HDD65	2794	672	583	474	205	37	3	0	0	38	252	529	
	CDD50	4923	3	10	67	259	515	660	908	991	768	509	203	30
	CDD65	1679	0	0	0	8	86	213	443	526	319	82	2	0
	CDH74	13060	0	0	0	11	185	822	4003	5593	2235	209	2	0
	CDH80	4061	0	0	0	0	13	119	1264	2089	555	21	0	0
Wind	WSAvg	11.6	11.4	11.9	12.6	12.8	12.1	11.0	11.8	11.9	11.5	11.1	10.4	10.7
Precipitation	PrecAvg	60.9	2.4	2.5	4.4	4.9	5.4	6.3	5.9	5.7	8.6	8.2	4.3	2.4
	PrecMax	80.3	5.6	5.6	7.8	9.2	10.1	12.6	12.8	14.0	18.2	26.9	10.1	7.8
	PrecMin	45.0	0.1	0.2	1.4	1.7	2.4	1.7	1.2	0.6	1.8	0.7	0.3	0.0
	PrecStd	8.6	1.6	1.4	1.5	2.0	2.0	2.3	3.6	3.7	4.1	5.8	2.4	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.7	64.6	69.4	75.4	82.1	86.4	93.2	94.9	90.0	82.6	73.0	66.3
		MCWB	52.9	55.9	57.7	63.1	66.8	74.2	78.0	79.6	76.8	71.7	63.1	57.6
	2%	DB	55.5	59.0	64.8	71.9	78.4	82.8	90.0	91.7	87.8	78.8	69.4	60.9
		MCWB	47.2	50.8	55.6	60.8	66.3	72.1	78.1	79.4	76.8	70.2	61.0	52.3
	5%	DB	52.1	54.0	62.4	69.7	75.5	80.7	88.1	89.9	85.7	75.6	66.4	57.5
		MCWB	43.5	46.0	54.5	60.6	65.3	71.4	77.8	78.5	76.1	67.8	58.9	49.4
	10%	DB	50.1	51.7	59.2	67.6	73.6	78.7	86.2	88.1	82.8	73.5	64.4	55.4
		MCWB	41.9	43.8	51.3	58.9	64.3	70.6	77.2	78.3	74.9	66.3	57.3	47.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	58.6	60.6	67.3	71.6	76.5	81.0	82.4	79.9	74.7	66.4	60.3
		MCDB	58.5	63.5	64.6	71.2	76.0	83.6	88.4	89.3	86.2	79.0	70.6	64.6
	2%	WB	48.6	52.1	57.9	64.1	69.3	74.7	79.8	81.0	78.5	72.2	62.9	54.3
		MCDB	53.9	57.5	63.4	69.4	74.6	80.4	86.6	87.6	84.1	77.1	66.8	59.3
	5%	WB	45.2	48.1	55.5	61.9	67.8	73.3	78.9	80.0	77.4	69.9	60.6	51.0
		MCDB	50.6	53.5	61.1	67.5	73.3	78.4	85.3	86.4	82.9	74.3	64.8	56.0
	10%	WB	42.9	45.0	52.2	59.9	66.3	72.0	78.0	79.3	76.4	67.5	58.7	48.8
		MCDB	48.8	49.9	58.3	65.5	71.8	76.7	84.3	85.5	81.8	72.3	63.2	54.5
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.5	12.2	12.3	11.3	9.7	10.0	10.3	9.0	9.0	10.1	11.1
		MCDBR	15.2	15.7	16.2	15.2	13.8	12.6	12.5	12.2	11.4	11.9	12.8	14.7
		MCWBR	11.8	12.8	11.8	9.6	7.6	6.3	5.2	4.9	5.5	8.3	10.1	11.5
	5% WB	MCDBR	13.8	14.9	15.0	12.8	12.0	10.7	11.4	11.3	10.5	10.7	11.5	13.2
MCWBR		12.0	13.3	12.6	9.5	8.2	5.9	5.1	5.0	5.7	8.5	10.5	11.8	
Clear-Sky Solar Irradiance	taub	0.383	0.432	0.499	0.549	0.570	0.602	0.594	0.581	0.517	0.488	0.430	0.385	
	taud	2.237	2.094	1.936	1.860	1.865	1.857	1.922	1.944	2.090	2.096	2.171	2.250	
	Ebn at Noon	251	251	244	239	236	228	228	228	237	230	232	241	
	Edh at Noon	34	44	56	64	64	65	60	58	48	44	36	32	
All-Sky Solar Radiation	RadAvg	884	1038	1296	1512	1624	1428	1570	1582	1220	960	795	771	
	RadStd	63	107	83	151	171	135	267	147	110	102	76	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page